

Merge K sorted Array



A man with dark hair and a mustache, wearing a dark blue patterned button-down shirt, stands on the left side of the frame. He is looking towards the camera. To his right is a chalkboard with a 4x4 grid of numbers. Each row of the grid is preceded by a checkmark (✓) written on the chalkboard. The numbers in the grid are: Row 1: 3, 4, 5, 11; Row 2: 1, 6, 7, 14; Row 3: 4, 5, 8, 13; Row 4: 6, 9, 10, 12. The grid is outlined in green.

✓	3	4	5	11
✓	1	6	7	14
✓	4	5	8	13
✓	6	9	10	12

Merge K sorted Array

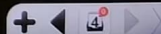
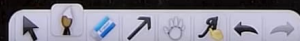
272 222



$K=4$

✓	3	4	5	11
✓	1	6	7	14
✓	4	5	8	13
✓	6	9	10	12

1	3	4	4	5	5	6		
---	---	---	---	---	---	---	--	--



Merge K sorted Array



$K = 4$

Merge sort

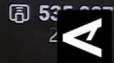
Heap-Sort

✓	3	4	5	11
✓	1	6	7	14
✓	4	5	8	13
✓	6	9	10	12

k^2

3	4	5	11	1	6	7	14	9	-	-	-
---	---	---	----	---	---	---	----	---	---	---	---

Merge K sorted Array



✓	3	4	5	11
✓	1	6	7	14
✓	4	5	8	13
✓	6	9	10	12

k^2

$K=4$

Merge sort

$n \log n$

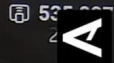
$k^2 \log k^2$

Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$

3	4	5	11	1	6	7	14	9	-	-	-
---	---	---	----	---	---	---	----	---	---	---	---

Merge K sorted Array



✓	3	4	5	11
✓	1	6	7	14
✓	4	5	8	13
✓	6	9	10	12

$K=4$

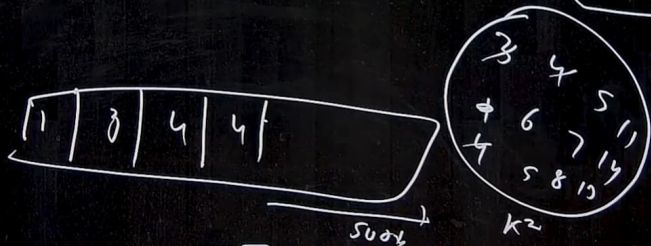
Merge Sort

$n \log n$

$k^2 \log k^2$

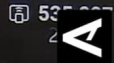
Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$



$k^2 \log k$ $O(k^2)$
 $k^2 \log k^2$
 $k^2 \log k^2$

Merge K sorted Array



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✓	1	6	7	14
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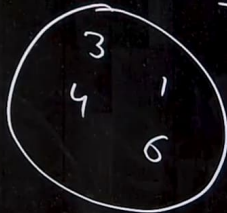
Merge sort

$n \log n$

$k^2 \log k^2$

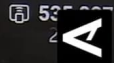
Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$



$k^2 \log k$, $O(k^2)$
 $k^2 \log k^2$
 $k^2 \log k^2$

Merge K sorted Array



✓	3 →	4	5	11
✓	1 →	6	7	14
✓	4 →	5	8	13
✓	6	9	10	12

$K = 4$

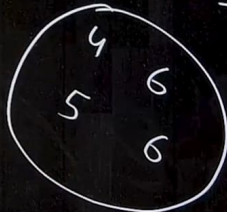
Merge sort

$n \log n$

$k^2 \log k^2$

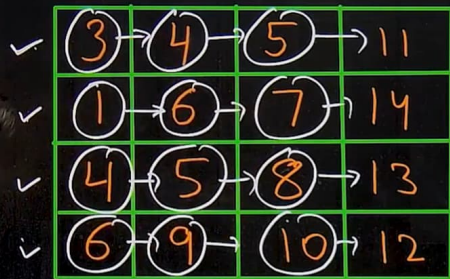
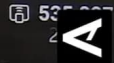
Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$



$k^2 \log k$ $O(k^2)$
 $k^2 \log k^2$
 $k^2 \log k^2$

Merge K sorted Array



$K = 4$

Merge sort

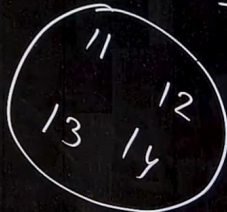
$n \log n$

$k^2 \log k^2$

Heap-Sort

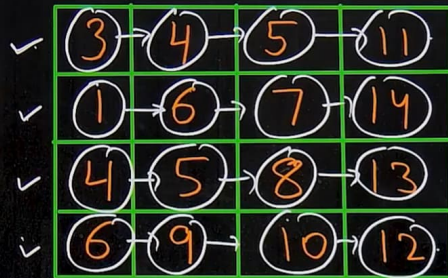
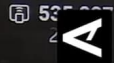
$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$

3 4 4 5 5 6 6
8 9 10



$k^2 \log k + O(k^2)$
 $k^2 \log k^2$
 $k^2 \log k^2$

Merge K sorted Array



$K=4$

Merge sort

$n \log n$

$k^2 \log k^2$

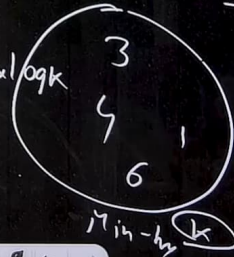
Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$

Delete

$$k^2 \log k + k + (k^2 - k) \times \log k$$

$$= k^2 \log k$$

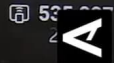


$$\frac{k^2 \log k}{k^2 \log k^2} = O(k^2)$$

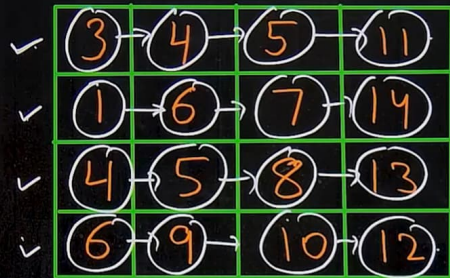
$$k^2 \log k^2$$

$$+ k^2 \log k^2$$

Merge K sorted Array



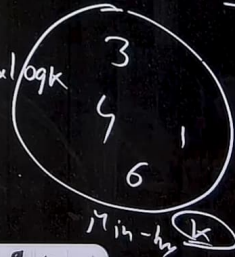
$\log n$
 k



Delete

$$k^2 \log k + k + (n^2 - k) \times \log k$$

Time $\frac{k^2 \log k}{k}$
Space $\frac{k}{k}$



$K=4$

Merge sort

$n \log n$

$k^2 \log k$

Heap-Sort

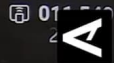
$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$

$\frac{k^2 \log k}{k} = O(k^2)$

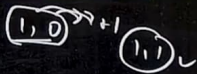
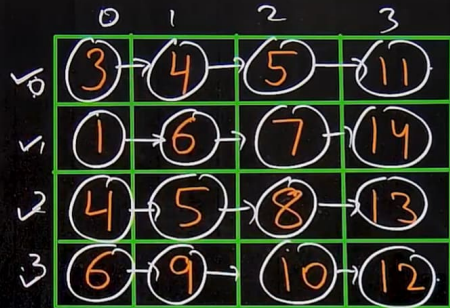
$k^2 \log k^2$

$k^2 \log k^2$

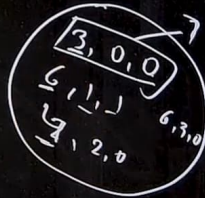
Merge K sorted Array



$K \log n$
 K



①



$K = 4$

Merge sort
 $n \log n$
 $K^2 \log K$

Heap-Sort

$K^2 + K^2 \log K$ $O(K^2 \log K)$
 $O(K^2 \log K)$ $O(1)$
 $O(K^2)$

Pair K_i : $K^2 \log K$ & $O(K^2)$
 For $K^2 \log K^2$
 $K^2 \log K^2$

Merge K sorted Array



	0	1	2	3
0	3	4	5	11
1	1	6	7	14
2	4	5	8	13
3	6	9	10	12

$K=4$

Merge sort

$n \log n$

$k^2 \log k$

Heap-Sort

$k^2 + k^2 \log k$ $O(k^2 \log k)$
 $O(k^2 \log k)$ $O(1)$
 $O(k^2)$

1. class Node
 {
 int value;
 int r;
 int col;
 }

3, 0, 0
 6, 1, 1
 9, 2, 0

Pair ki $k^2 \log k$ & $O(k^2)$
 For $k^2 \log k^2$
 $k^2 \log k^2$



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Editorial

🕒 Submissions

💬 Comments

C++ (g++ 5.4)

Average Time: 45m

🕒 Start Timer



Merge k Sorted Arrays



Medium

Accuracy: 67.25%

Submissions: 84K+

Points: 4

Given K sorted arrays arranged in the form of a matrix of size K*K. The task is to merge them into one sorted array.

Example 1:

Input:

K = 3

arr[] = {{1,2,3},{4,5,6},{7,8,9}}

Output: 1 2 3 4 5 6 7 8 9

Explanation: Above test case has 3 sorted arrays of size 3, 3, 3

arr[] = [[1, 2, 3],[4, 5, 6],
[7, 8, 9]]

The merged list will be

[1, 2, 3, 4, 5, 6, 7, 8, 9].

Example 2:

Input:

K = 4

arr[]={{1,2,3,4},{2,2,3,4},
{5,5,6,6},{7,8,9,9}}

Output:

1 2 2 2 3 3 4 4 4 5 5 6 6 7 8 9 9

```
1 // } Driver Code Ends
15 //User function Template for C++
16
17
18 class Solution
19 {
20     public:
21         //Function to merge k sorted arrays.
22
23
24         vector<int> mergeKArrays(vector<vector<int>> arr, int n)
25         {
26             //code here
27
28
29         }
30 };
31 // } Driver Code Ends
```



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Example 2:

Input:

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Output:

1 2 2 2 3 3 4 4 5 5 6 6 7 8 9 9

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24         vector<int> mergeKArrays(vector<vector<int>> arr, int n)
25         {
26             //code here
27             pair<int,pair<int,int>>|
28             // Min heap
29             priority_queue<int, vector<int>, greater<int>>p;
30             // data, row, col (pair)
31         }
32     };
33 // } Driver Code Ends
```



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Example 2:

Input:

K = 4

arr[]={{1,2,3,4},{2,2,3,4},
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Output:

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Solution

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23
24 vector<int> mergeKArrays(vector<vector<int>> arr, int K)
25
26 //code here
27 vector<pair<int,pair<int,int>>>temp;
28 // Min heap
29 priority_queue<pair<int,pair<int,int>>, vector<pair<int,pair<int,int>>>, greater<pair<int,pair<int,int>>>>temp;
30 // data, row, col (pair)
31
32
33 } Driver Code Ends
```



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Output:

1 2 2 2 3 3 4 4 5 5 6 6 7 8 9 9

```
1 } Driver Code Ends
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```

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```
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19
20 public:
21 //Function to merge k sorted arrays.
22
23
24 vector<int> mergeKArrays(vector<vector<int>> arr, int K)
25
26 //code here
27 vector<pair<int,pair<int,int>>>temp;
28 for(int i=0;i<K;i++)
29 {
30     temp.push_back(make_pair(arr[i][0], make_pair(i,0))
31 }
32 // Min heap
33 priority_queue<pair<int,pair<int,int>>, vector<pair<int,pair<int,int>>>>temp>>
34 // data, row, col (pair)
```



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The merged list will be

[1, 2, 3, 4, 5, 6, 7, 8, 9].

Example 2:

Input:

K = 4

arr[][] = {{1,2,3,4},{2,2,3,4},
 {5,5,6,6},{7,8,9,9}}

Output:

1 2 2 2 3 3 4 4 5 5 6 6 7 8 9 9

17
18
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25
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36
37

```
, greater<pair<int,pair<int,int>>>>p(temp.begin(),temp.e
```

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Output:

1 2 2 2 3 3 4 4 5 5 6 6 7 8 9 9

```
23
24 vector<int> mergeKArrays(vector<vector<int>>> arr, int n)
25 {
26     //code here
27     vector<pair<int,pair<int,int>>>temp;
28     for(int i=0;i<K;i++)
29     {
30         temp.push_back(make_pair(arr[i][0], make_pair(i,0)));
31     }
32     // Min heap
33     priority_queue<pair<int,pair<int,int>>, vector<pair<int,pair<int,int>>>,
34     // data, row, col (pair)
35     greater<pair<int,pair<int,int>>>>temp;
36     vector<int>ans; // vector which will store the sorted array
37     pair<int,pair<int,int>>Element;
38     int i, int j;
39
40     while(!temp.empty())
41     {
42         Element = temp.top(); temp.pop();
43         // } Driver Code Ends
```



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Output:

1 2 2 2 3 3 4 4 5 5 6 6 7 8 9 9

```
31 }
32 // Min heap
33 priority_queue<pair<int,pair<int,int>>, vector<pair<int,int>>> p;
34 // data, row, col (pair)
35
36 vector<int>ans; // vector which will store the sorted array
37 pair<int,pair<int,int>>Element;
38 int i, int j;
39
40 while(!p.empty())
41 {
42     Element = p.top();
43     p.pop();
44     ans.push_back(Element.first);
45     i = Element.second.first;
46     j = Element.second.second;
47     if(j+1<K)
48         p.push(make_pair(arr[i][j+1],make_pair(i,j+1)));
49 }
50
51 return ans;
```



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